Pomona

Maya Holloway

Technical Field

While the main focus of this project is artificial intelligence, it does have limited aspects of embedded systems.

Background Information

With a focus on localized processing and data storage, the idea for Pomona stems from a desire to provide powerful data insights without dependence on outside resources or providers. Agentic oversight and predetermined flags are now standard for small and big businesses alike, but small farms and rural homesteads often have limited data management or analysis. Focused, automated data insights can multiply the force of small groups of workers and owners exponentially with limited time or input.

Prior Art (Research)

Loom:
The product of a team of scientists from Oregon, Loom is a template for a datalogging system related to environmental sensors for research and ecological survey. A wonderful and inspirational tool, Loom does recommend an engineering background for use and is seemingly very code/software intensive.
Richards, W., Selker, J., & Udell, C. (2024). Loom: A Modular Open-Source Approach to Rapidly Produce Sensor, Actuator, Datalogger Systems. Sensors, 24(11), 3466. https://doi.org/10.3390/s24113466

Trackpac LoRaWAN Smart Agriculture:
This is a paid service that focuses on customizable smart agriculture. The data generated is shared with the company and the subscription cost grows with every sensor addition. This doesn’t seem to focus on sustainable power solutions.

IoT Sensor Management Made Easy - Trackpac. (n.d.). LoRaWAN smart agriculture sensors | Farm IoT monitoring | Trackpac. https://trackpac.io/sensors/lorawan-smart-agriculture/

Monnit Agriculture and Livestock Remote Monitoring:
This is another paid service that focuses on customizable smart agriculture. The data generated is

shared with the company as well as a cloud provider and the subscription cost also grows with every sensor addition. This doesn’t seem to focus on sustainable power solutions either.

Monnit. (n.d.). Agriculture and livestock remote monitoring solutions. https://www.monnit.com/applications/agriculture-livestock-monitoring/?srsltid=AfmBOoroF_iB1at-77l269x5_bM9CTIf3wFAC19TWH-SmRuguFoTTIdn

Project Description

Pomona focuses on energy efficiency, sustainable power generation and limiting the required time and input of the user while still being both powerful, modular and reliable. This project will provide a low or no cost solution to data collection and analysis for small farm owners and homesteaders in rural areas by ensuring compatibility with off grid power solutions and ease of use. With simple, focused agentic oversight powered by a local LLM, Pomona provides an invaluable service to those that want to maximize their efficiency and output without compromising data security.

Innovation Claim

A simple edge processor with a single sensor compartmentalizes small data packets for translation and storage in a local database, allowing a local LLM powered agent to identify patterns before they become costly issues or repairs.

Usage Scenario

While a single sensor only offers so much information to be gleaned, later iterations can be expanded to multiple nodes, multiple sensors, and more advanced energy solutions; with more data integrated into predetermined dashboards that allow customization, farmers and homesteaders gain access to further insights and potential issues. From fire alarms to water levels, Pomona is limited only be the sensors available and the imagination of the user.